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Catalase (CAT) activity assay for zooplankton samples

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Engström-Öst, J., Glippa, O., Feely, R.A. et al. Eco-physiological responses of copepods and pteropods to ocean warming and acidification. Sci Rep 9, 4748 (2019). <https://doi.org/10.1038/s41598-019-41213-1>

Jonna Engström-Öst, Mirella Kanerva, Kristiina A. Vuori, Ulf Riebesell, Carsten Spisla, Olivier Glippa, Oxidative stress and antioxidant defence responses in two marine copepods in a high CO₂ experiment, Science of The Total Environment, Volume 745, 2020, <https://doi.org/10.1016/j.scitotenv.2020.140600>.

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Abstract

Catalase (CAT) activity measurement from copepod (*Limnocalanus macrurus*, *Acartia* sp., *Eurytemora affinis*, *Temora longicornis*, *Calanus* sp.), cladoceran (*Eubosmina maritima*) and pteropod (*Limacina helicina*) homogenates. This measurement for catalase (CAT) activity is a modified microplate version of the colorimetric catalase assay, where a reaction of catalase and H_2O_2 is stopped with NaN_3 (Deissero & Dounce, 1970), an aliquot is pipetted to a microplate and the leftover H_2O_2 is detected with colorimetric reaction (Fossati *et al.*, 1980).

Deissero.A & Dounce, A. L. (1970). Catalase - physical and chemical properties, mechanism of catalysis, and physiological role. *Physiological Reviews* 50, 319-375.

Fossati, P., Prencipe, L. & Berti, G. (1980). Use of 3,5-dichloro-2-hydrobenzenesulfonic acid-4-aminophenazone chromogenic system in direct enzymic assay of uric-acid in serum and urine. *Clinical Chemistry* 26, 227-231.

Guidelines

Samples and sample dilutions should be kept on ice. The reactions are done at RT.

Materials

MATERIALS

☒ EnSpire or EnVision Multilabel Plate Reader **Perkin Elmer**

☒ 96-well plate, transparent **Perkin Elmer**

☒ Single channel and multichannel pipettes

☒ CAT Assay buffer 10X (500 mM potassium phosphate buffer, pH 7.0)

☒ Stop Solution (15 mM sodium azide in water)

☒ Sample Dilution Buffer (50 mM potassium phosphate buffer, pH 7.0, containing 0.1% TRITON X-100)

☒ Substrate Solution (200 mM hydrogen peroxide)

☒ Standard Solution (10 mM hydrogen peroxide)

☒ Peroxidase from horseradish **Merck MilliporeSigma (Sigma-Aldrich) Catalog #P6782**

☒ Color Reagent (150 mM potassium phosphate buffer, pH 7.0, containing 0.25 mM 4-aminoantipyrine and 2 mM 3,5-dichloro-2-hydroxybenzenesulfonic acid)

Safety warnings

! See reagents' MSDSs.

Before start

Dilute 10x Assay Buffer to 1x Assay Buffer and equilibrate at RT.

H₂O₂: Determine the exact concentration by UV absorbance and make fresh dilutions (200 mM and 10 mM)

accordingly every day. Equilibrate at RT.

Peroxidase: Weigh 1 mg of solid powder and dissolve in 1.45 ml of 1X Assay Buffer. The solution can be stored at 4 °C for up to 2 weeks.

Equilibrate Stop Solution at RT.

Prepare the Colour Reagent by adding 30 µl of the Peroxidase Solution to each 30 ml of Colour Reagent. Equilibrate at RT.

Dilute samples to 1 mg/ml protein with sample dilution buffer. If the samples have less than 1mg/ml protein, the dilutions are made directly to the reaction tube.

Prepare standard from 10 mM H₂O₂: 0, 0.0025, 0.005, 0.01, 0.015 mmol



Buffer and sample

- 1 Pipette to Eppendorf tubes:
13 μl 1x Assay Buffer / Blank: 15 μl 1x Assay Buffer
2 μl sample (see before starting section for sample protein concentration) / Blank: no sample

Start reaction

- 2 Start catalase reaction with 5 μl of 200 mM H_2O_2 , incubate 5 min. Note: Make sure reaction time is the same for all samples and is accurately known, since it is important for activity calculation.

Stop reaction

- 3 Stop catalase reaction with 180 μl Stop solution

Reaction mixture and colour reagent to the 96-well plate

- 4 Within 15 min pipette 2 μl of catalase reaction mixture in wells (96 well plate) and add 200 μl of colour reagent (for colorimetric reaction). The standards (2 μl /well) are pipetted directly to the plate at this point and 200 μl of colour reagent added as for the samples.

Measure absorbance

- 5 Measure absorbance at 520 nm after 15 min.

Calculate results

- 6 Determine the amount of H_2O_2 (μmoles) remaining in the Colorimetric Reaction mixture using the H_2O_2 standard curve:
 $A_{520}(\text{Blank}) = \mu\text{moles of } \text{H}_2\text{O}_2 \text{ in Blank}$
 $A_{520}(\text{Sample}) = \mu\text{moles of } \text{H}_2\text{O}_2 \text{ in Sample}$
 $\Delta\mu\text{moles } (\text{H}_2\text{O}_2) = \mu\text{moles of } \text{H}_2\text{O}_2 (\text{Blank}) - \mu\text{moles of } \text{H}_2\text{O}_2 (\text{Sample})$

Calculate the activity in samples:



Catalase (CAT) activity ($\mu\text{moles}/\text{min}/\text{ml}$) = $(\Delta\mu\text{moles (H}_2\text{O}_2) \times d \times 100) / (V \times t)$

$\Delta\mu\text{moles (H}_2\text{O}_2) = \mu\text{moles of H}_2\text{O}_2 \text{ (Blank)} - \mu\text{moles of H}_2\text{O}_2 \text{ (Sample)}$

d = dilution of original sample for catalase reaction

t = catalase reaction duration (minutes)

V = sample volume in catalase reaction ($x \mu\text{l} = 0.00x \text{ ml}$)

100 = dilution of aliquot from catalase reaction in colorimetric reaction (2 μl in 0.2 ml)